

Autonomous Cognitive Engine for Deep Research and Long- Horizon Tasks

Presented by,
Divyadharshini.S

Overview:

- AI-based system for complex, long-running research workflows
- Designed to maintain context, planning, and execution over time
- Uses a multi-agent, state-driven architecture
- Automates research-oriented tasks such as:
 - Information retrieval
 - Summarization
 - Task tracking
 - Memory management

Problem Statement:

Why This Project?

- Traditional LLM systems:
 - Lose context in long conversations
 - Lack structured planning
 - Produce inconsistent outputs
- Manual research workflows are:
 - Time-consuming
 - Difficult to track
 - Error-prone

Approach & System Architecture:

- Agent-Driven Modular Approach
 - System follows a Supervisor–Worker architecture
 - Each agent has a clear, dedicated responsibility
 - Execution is state-driven and sequential
 - Shared state ensures context continuity
- Core Components
 - Supervisor Agent
 - Worker Agent
 - Virtual File System (VFS)
 - LangGraph-based workflow orchestration

Agents :

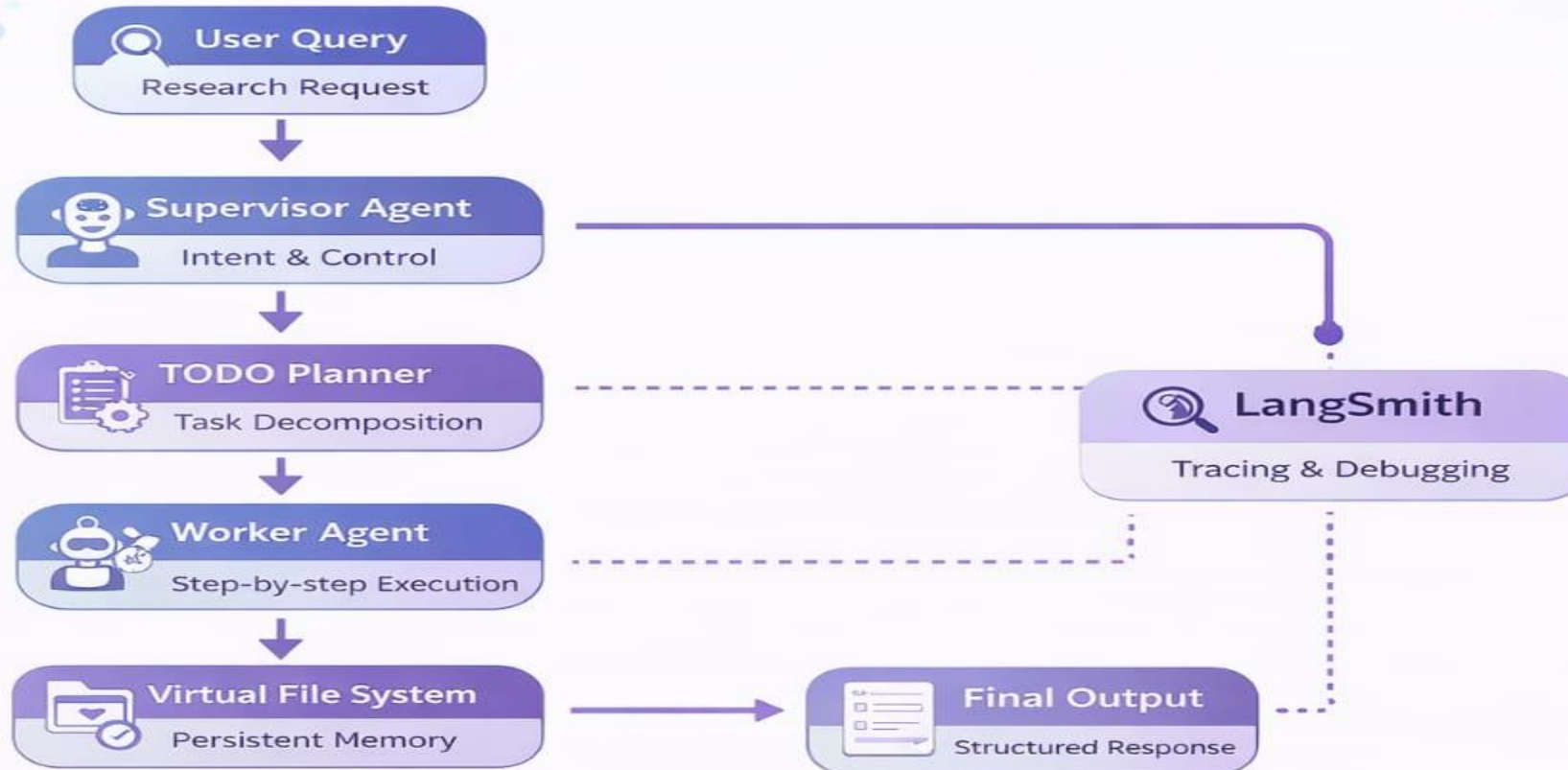
Agent Responsibilities

- **Supervisor Agent**
 - Interprets user intent
 - Generates structured TODO plans
 - Controls execution flow
- **Worker Agent**
 - Executes tasks step-by-step
 - Stores intermediate outputs
 - Updates shared state

Workflow Execution:

Execution Flow of Autonomous Cognitive Agent

A state-driven, explainable workflow for long-horizon tasks



Technology Stack:

Tools & Technologies Used

- Programming Language:
- Python 3.10+
- Agent Frameworks: LangChain, LangGraph
- LLM Integration: Groq API (LLaMA 3.1)
- Environment Management: python-dotenv
- Observability: LangSmith
- User Interface: Streamlit
- Version Control: Git & GitHub (branch-based workflow)

User Interface (Streamlit UI):

Interactive User Experience

- Chat-based interface
- Real-time agent responses
- Persistent session state
- Clear visualization of task execution
- UI Advantage
 - Easy interaction with autonomous agents
 - Suitable for both research and automation use cases

LangSmith Observability & Debugging

LangSmith Integration

- Tracks agent execution paths
- Visualizes:
 - Task decomposition
 - LLM calls
 - State transitions
- Helps in:
 - Debugging recursion issues
 - Improving prompt design
 - Understanding agent behavior
- Benefit
 - Makes the system transparent and explainable

 LangSmith 

Application

 All applications 

 Home

 Tracing 9

 Monitoring

 Datasets & Experiments

 Annotation Queues

 Prompts

 Playground

 Studio

 Deployments

 Documentation

 Invitations

 Theme

 Settings

 Personal

sdivyadharshinisaravan...

Personal > Tracing >  milestone3subagent

 **milestone3subagent**  ID **Runs**

1 filter Last 7 days Traces LLM Calls

☐		Name	Input
☐		Main Agent Run	plan a b
☐		Main Agent Run	plan a 6
☐		Main Agent Run	plan a f
☐		Main Agent Run	plan a f
☐		Main Agent Run	Design
☐		Main Agent Run	summa
☐		Main Agent Run	rovide
☐		Main Agent Run	plan a c
☐		Main Agent Run	summa
☐		Main Agent Run	provide

TRACE

Waterfall

Main Agent Run 3.64s 1.3K

- Supervisor Agent 3.56s 1,291
 - Planner Agent 0.00s
 - Planner Tool 0.00s
 - Search Agent 1.69s 609
 - Search Tool 1.69s 609
 - Chat... llama-3.1-8... 1.69s 609
 - Summarizer Agent 1.26s 682
 - Summarizer Tool 1.26s 682
 - Chat... llama-3.1-8...

Compare

Main Agent Run

ID

Playground

+

↑

↻

Run

Feedback

Metadata

Input ▾

▾ User Input

plan a birthday party for my mom and provide some websites for it and also summarize it all

▾ State

> Current Task plan a birthday party...

> Final Answer **Party Plan for You...

> Messages [...]

▾ Todo Status

▾ 0

> Agent Planner

> Output Skipped

> Status Skipped

> Task plan a birthday party...

> Tool Planner Tool

▾ 1

Collapse

START TIME

01/20/2026, 06:53:36 PM GMT+5

END TIME

01/20/2026, 06:53:40 PM GMT+5

Status

✓ Success

TOTAL TOKENS

1,291 tokens

LATENCY

🕒 3.64s

Type

Chain

Polly

Challenges Faced:

- Managing long-term context within LLM limits
- State recursion errors in LangGraph workflows
- Coordinating shared state across multiple agents

Future Scope:

- Parallel execution using multiple worker agents
- Distributed cloud-based memory
- Integration with external tools (web search, APIs)
- Advanced task prioritization and scheduling

Conclusion:

- Demonstrates a scalable autonomous agent system
- Effectively combines:
 - Planning
 - Reasoning
 - Execution
 - Persistent memory
- Provides a strong foundation for advanced AI research and automation



Q & A



THANK YOU!